Computing Mathematics
BBC Finals
PROSSIE'S PHOTOCOPIER

MAKERERE UNIVERSITY BUSINESS SCHOOL
END OF SEMESTER EXAMINATION FOR THE
DEGREE OF BACHELOR OF BUSINESS COMPUTING (BBC)
OF MAKERERE UNIVERSITY, ACADEMIC YEAR 2022/ 2023

COURSE NAME: Computing Mathematics
COURSE CODE: BUC1114
YEAR OF STUDY: One
SEMESTER: One
DATE: Sunday 29th January 2023
TIME: 9.00am –12:00Noon

INSTRUCTIONS

[1]. Time allowed is 3 hours

[2]. Attempt Section A and **ANY THREE** Questions from section B.

[3]. Use of relative examples and illustrations will earn you credit.

Section A (40 Marks)

1. Convert the decimal number 578.45 to binary. *(04 marks)*
2. Decode the number 7CA2H into decimal values. *(04 marks)*
3. What distinguishes ASCII from EBCDIC alphanumeric codes. *(04 marks)*
4. Express the number 8579_{10} in Binary Coded Decimal. *(04 marks)*
5. Use doubling the dabble to change the binary number 1111 0000 1100 10101 to decimal values. *(04 marks)*
6. Given the functions: $f(x) = x^2 + 4x - 1$ *and* $g(x) = 5 - 2x$. Find $f(4) + g(4)$. *(04 marks)*
7. Using a Venn diagram, show that $(A \cap B)' = A' \cup B'$. *(04 marks)*
8. Write a pseudo code for a program that tells a user to enter a number and checks if it is even or odd. *(04 marks)*
9. Define a matrix and give an example. *(04 marks)*
10. Distinguish between an integer and a floating-point number. *(04 marks)*

Section B (60 Marks)

Question 11.

a. What is the basic unit of data representation in a computer? *(01 mark)*

b. Define what is meant by the term radix. *(01 mark)*

c. What are the most commonly used radices for data representation in computers. *(04 marks)*

d. Give an example in computing where each of the named radices in (c) above are used. *(04 marks)*

e. Explain how negative and positive numbers are represented using signed magnitude representation. *(06 marks)*

f. List down the techniques of detecting and correcting errors in data representation in computers *(04 marks)*

Question 12:

a. Solve the system of equations $3x - 2y = 8$ and $x + 4y = -3$ using
   i. Inverse matrix method (07 *marks*)
   ii. Row operations matrix (08 *marks*)

b. Give examples in computing where systems of equations and matrices are used. (*05 marks*)

Question 13:

a. Define a function and its inverse. (*04 marks*)

b. Given the function: $f(x) = 2x^2 + x - 5$
   a. Plot sketch graph of $f(x)$ (*05 marks*)
   b. Find the inverse of $f(x)$. (*06 marks*)

c. List down applications of functions and their inverses in computing (*05 marks*)

Question 14:

a. Given that U= {1,2,3,4,5,6,7,8,9,10} $A = \{1,2,3,4,5,6,7\}$ $B = \{2,4,6,,8,10\}$. Find

(i)	$A' \cap B$	(02 *marks*)
(ii)	$A \cup B'$	(02 *marks*)
(iii)	$n(A' \cap B')$	(02 *marks*)

b. Fifty (50) university students were asked whether they like Adventure stories, Detective novels or Romantic fiction. 8 like Adventure stories and Detective novels, 11 like Romantic fiction and Detective novels and 19 like Adventure stories and Romantic fiction. 6 like Romantic fiction only, 7 like adventure stories only and 8 like detective novels only.
   (i) How many like all three, if three of the students like none of them? (*06 marks*)
   (ii) How many students like at most two? (*03 marks*)

c. Give applications of sets in computing (*05 marks*)

Question 15:

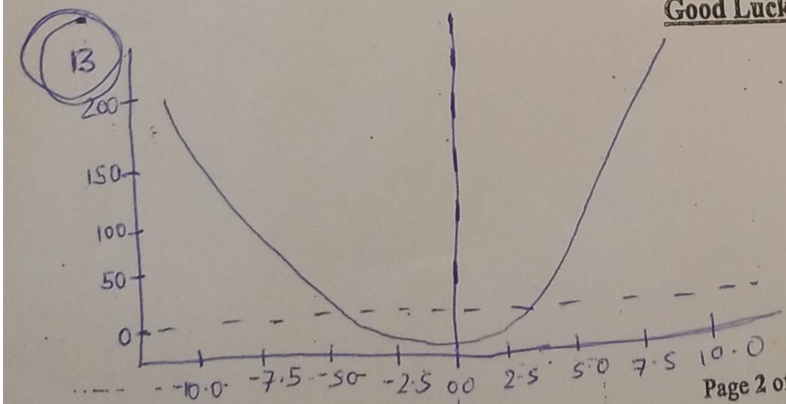
a. Define an algorithm. (*02 marks*)

b. Give three properties of an algorithm. (*04 marks*)

c. Explain the three ways of expressing algorithms. (*06 marks*)

d. Describe the components of a pseudo-code? (*08 marks*)

Good Luck



The plot shows the parabola of the given quadratic function, which opens upwards as the coefficient of x^2 is positive.